Java LinkedHashSet

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Java LinkedHashSet

A `LinkedHashSet` is a collection that stores **unique elements** and **remembers the order they were added**.

It is part of the `java.util` package and implements the `Set` interface.

Tip: Use `LinkedHashSet` when you want a set that does not allow duplicates *and* keeps the original insertion order.

Create a LinkedHashSet

Example

Create a `LinkedHashSet` object called **cars** that will store strings:

```
import java.util.LinkedHashSet; // Import the LinkedHashSet class
```

Now you can use methods like `add()`, `contains()`, and `remove()` to manage your collection.

Add Elements

To add elements to a `LinkedHashSet`, use the `add()` method:

Example

```
import java.util.LinkedHashSet;

public class Main {
  public static void main(String[] args) {
    LinkedHashSet<String> cars = new LinkedHashSet<>();
    cars.add("Volvo");
    cars.add("BMW");
    cars.add("Ford");
    cars.add("BMW");  // Duplicate
    cars.add("Mazda");

    System.out.println(cars);
  }
}
```

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Output: The elements will appear in the order they were added (e.g., [Volvo, BMW, Ford, Mazda]).

Check if an Element Exists

Use `contains()` to check for an element:

Example

```
cars.contains("Mazda");
```

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Remove an Element

Use `remove()` to remove an element:

Example

```
cars.remove("Volvo");
```

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Remove All Elements

Use `clear()` to remove all elements:

```
cars.clear();
```

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LinkedHashSet Size

Use `size()` to count how many unique elements are in the set:

Example

```
cars.size();
```

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Note: Duplicate values are not counted - only unique elements are included in the size.

Loop Through a LinkedHashSet

Loop through the elements of a `LinkedHashSet` with a **for-each** loop:

Example

```
LinkedHashSet<String> cars = new LinkedHashSet<>();
// Add elements...

for (String car : cars) {
```

HashSet vs LinkedHashSet

Feature	`HashSet`	`LinkedHashSet`
Order	No guaranteed order	Insertion order preserved
Duplicates	Not allowed	Not allowed
Performance	Faster	Slightly slower (due to order tracking)

Tip: Use `HashSet` when you only care about uniqueness and speed. Use `LinkedHashSet` when order matters.

The var Keyword

From Java 10, you can use the `var` keyword to declare a `LinkedHashSet` variable without writing the type twice. The compiler figures out the type from the value you assign.

This makes code shorter, **but many developers still use the full type for clarity**. Since `var` is valid Java, you may see it in other code, so it's good to know that it exists:

Example

```
// Without var
LinkedHashSet<String> cars = new LinkedHashSet<String>();
```

The Set Interface

Note: Sometimes you will see both `Set` and `LinkedHashSet` in Java code, like this:

```
import java.util.Set;
import java.util.LinkedHashSet;

Set<String> cars = new LinkedHashSet<>();
```

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This means the variable (cars) is declared as a `Set` (the interface), but it stores a `LinkedHashSet` object (the actual set). Since `LinkedHashSet` implements the `Set` interface, this is possible.

It works the same way, but some developers prefer this style because it gives them more flexibility to change the type later.

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